



ADAM  
SHAW

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## PROFESSIONAL SUMMARY

Third-year undergraduate at USC pursuing a B.S. in Applied Mathematics (CS minor) and M.S. in Statistics, with software engineering experience at Google and research experience in topological deep learning. Seeking PhD opportunities in Machine Learning to continue research in neural network learning dynamics.

## SKILLS

- Python
- Scikit-learn
- PyTorch
- Pandas
- JAX
- C++

## EDUCATION

**University of Southern California** | Los Angeles, CA

Expected to Graduate December 2027

B.S., Applied Mathematics; Computer Science Minor

M.S., Statistics

Cumulative GPA: 3.7/4.0

Relevant Coursework (All A's):

Graduate Analysis of Algorithms, Mathematics of Machine Learning, Mathematical Statistics, Data Science with Python

## EXPERIENCE

Student Researcher

August 2025 - Present

**University of Southern California** | Los Angeles, CA

- Analyzing changes in topological invariants (Betti numbers) across network layers and their effect on inversion ability in [predictive coding networks](#), a type of invertible neural network
- Developed extensive Python codebase implementing predictive coding networks in JAX and topological analysis using Ripser and NumPy/Scikit-learn
- Presented findings at the [Conference on Topological Data Analysis](#) at the University of Missouri in November and preparing manuscript for submission

Associate Software Developer Intern

May 2025 - August 2025

**Google** | New York, NY

- Designed and built an access control system for custom Gems (personalized Gemini instances), streamlining creation and secure sharing across Google Workspace teams
- Implemented role-based permissions in C++ using Google's internal server framework, managing authorization and routing for millions of potential users
- Developed Python automation script to manage enum ID generation and updates across a 10,000+ line shared codebase, eliminating merge conflicts and synchronization errors for a multi-engineer team

Student Researcher

December 2024 - May 2025

**University of Southern California** | Los Angeles, CA

- Worked with Professor James Alcalá on the identification and classification of bias in text containing hyperbolic and sarcastic language
- Researched existing bias classification systems that quantify political/ideological bias, and looked at how these could be applied to unconventional text